To find mean of the distribution

x	f	fx
0	19	0
1	99	99
2	197	394
3	198	594
4	105	420
5	22	110
Total	640	1617

Mean : $\bar{x} = \frac{\sum fx}{\sum f} = \frac{1617}{640} = 2.526$

The probability mass function of binomial distribution is :

$p(x) = {}^nC_x\, p^x\, q^{n-x},\ x = 0,1,\ldots, n$ (2.1)

Mean of the binomial distribution is $\bar{x} = np$.

Hence, $\hat{p} = \frac{\bar{x}}{n} = \frac{2.526}{5} \approx 0.5$

$\hat{q} = 1 - \hat{p} \approx 0.5$

For $x = 0$, the equation (2.1) becomes

$P(X = 0) = P(0) = 5c_0\ (0.5)^5 = 0.03125$

The expected frequency at $x = N\,P(x)$

The expected frequency at x =0 : $N \times P(0)$

$= 640 \times 0.03125 = 20$

We use recurrence formula to find the other expected frequencies.

The expected frequency at x+1 is

$\frac{n-x}{x+1}\left(\frac{p}{q}\right) \times$ Expected frequency at x

x	$\frac{n-x}{x+1}$	$\frac{p}{q}$	$\frac{n-x}{x+1}\left(\frac{p}{q}\right)$	Expected frequency at $x = N\,P(x)$
0	5	1	5	20
1	2	1	2	100
2	1	1	1	200
3	0.5	1	0.5	200
4	0.2	1	0.2	100
5	0	1	0	20

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